



Oxidative burden of fine particulate air pollution and risk of cause-specific mortality in the Canadian Census Health and Environment Cohort (CanCHEC)

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ABSTRACT

Background: Fine particulate air pollution (PM_{2.5}) is known to contribute to cardiorespiratory mortality but it is not clear how PM_{2.5} oxidative burden (i.e. the ability of PM_{2.5} to cause oxidative stress) may influence long-term mortality risk.

Methods: We examined the relationship between PM_{2.5} oxidative burden and cause-specific mortality in Ontario, Canada. Integrated PM_{2.5} samples were collected from 30 provincial monitoring sites between 2012 and 2013. The oxidative potential (% depletion/μg) of regional PM_{2.5} was measured as the ability of filter extracts to deplete antioxidants (glutathione and ascorbate) in a synthetic respiratory tract lining fluid. PM_{2.5} oxidative burden was calculated as the product of PM_{2.5} mass concentrations and regional estimates of oxidative potential. In total, this study included 193,300 people who completed the Canadian long-form census in 1991 and who lived within 5 km of a site where oxidative potential was measured. Deaths occurring between 1991 and 2009 were identified through record linkages and Cox proportional hazard models were used to estimate hazard ratios (and 95% confidence intervals) for interquartile changes in exposure adjusting for individual-level covariates and indirect-adjustment for smoking and obesity.

Results: Glutathione-related oxidative burden was associated with cause-specific mortality. For lung cancer specifically, this metric was associated with a 12% (95% CI: 5.0–19) increased risk of mortality whereas a 5.0% (95% CI: 0.1, 10) increase was observed for PM_{2.5}. Indirect adjustment for smoking and obesity decreased the lung cancer hazard ratio for glutathione-related oxidative burden but it remained significantly elevated (HR=1.07, 95% CI: 1.005, 1.146). Ascorbate-related oxidative burden was not associated with mortality.

Conclusions: Our findings suggest that glutathione-related oxidative burden may be more strongly associated with lung cancer mortality than PM_{2.5} mass concentrations.

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1. Introduction

Fine particulate air pollution (PM_{2.5}) is known to contribute to cardiorespiratory morbidity and mortality and is recognized as an

Abbreviations: OP^{AA}, Ascorbate-related oxidative potential; OP^{GSH}, Glutathione-related oxidative potential

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important contributor to global disease burden (Lim et al., 2012). While the biological mechanisms underlying these associations have yet to be fully elucidated, existing evidence suggests that oxidative stress plays an important role in PM-induced health effects including systemic/respiratory inflammation and DNA damage (Araujo and Nel, 2009; Knappen et al., 2004; Li et al., 2003; Moller et al., 2014; Weichenthal et al., 2013). In addition, sources such as traffic and biomass burning are known to influence the oxidative potential of particulate air pollution (Bates et al., 2015; Boogaard et al., 2012; Godri et al., 2011; Janssen et al., 2014; Kelly

et al., 2011; Kurmi et al., 2013; Yang et al., 2015a; Yanosky et al., 2012) and the oxidative properties of PM_{2.5} vary between cities (Künzli et al., 2006). Nevertheless, exposure assessment methods in large population-based studies are rarely designed to capture spatial differences in the ability of PM_{2.5} to cause oxidative stress. In particular, large cohort studies have yet to examine the impact of PM_{2.5} oxidative burden on long-term mortality risk. Such studies are of interest as incorporating this parameter may result in stronger risk estimates if PM_{2.5} oxidative burden is more closely related to the biologically effective dose (Borm et al., 2007). Moreover, defining exposures in terms of PM_{2.5} oxidative burden may reduce heterogeneity between studies, as mass concentration measurements may not adequately capture regional differences in biological activity.

In this study, the term *oxidative potential* is used to reflect the ability of PM_{2.5} to deplete anti-oxidants in a synthetic respiratory tract lining fluid (% depletion/ μg PM_{2.5}). The term *oxidative burden* is used to reflect the product of PM_{2.5} mass concentrations and regional estimates of PM_{2.5} oxidative potential. The synthetic respiratory tract lining fluid used in this study contained the anti-oxidants glutathione and ascorbate as these compounds act as a first line of defense against inhaled pollutants (Kelly, 2003). Moreover, the ratio of reduced to oxidized glutathione (GSH/GSSG) is an important parameter in the regulation of airway inflammation (Rahman and MacNee, 2000). Our hypothesis was that PM_{2.5}-oxidative burden may be more strongly associated with mortality outcomes than traditional mass-based measurements as PM_{2.5} oxidative burden may provide a more accurate reflection of spatial differences in the ability of PM_{2.5} to cause oxidative stress.

2. Methods

2.1. Cohort description

The Canadian Census Health and Environment Cohort (CanCHEC) has been described in detail previously (Crouse et al., 2012; Peters et al., 2013). Briefly, CanCHEC is a population-based cohort of subjects who were 25 years of age or older at baseline, a usual resident of Canada on the Census reference day (June 4, 1991), not a resident of an institution such as a prison, hospital, or nursing home, and among the 20% of Canadian households (about 3.6 million respondents) selected for enumeration with the mandatory long-form Census questionnaire. As part of the census, subjects also provided detailed information on a number of individual-level socio-demographic variables (Table 1). Subjects in CanCHEC were linked to the Canadian Mortality Database using deterministic and probabilistic linkage methods from June 4, 1991 through December 31, 2009. Ascertainment of mortality in the cohort is estimated to be approximately 97% (Wilkins et al., 2008). The date of death and the underlying cause of death were extracted from death certificates coded to the International Classification of Diseases, 9th Revision (ICD-9 (WHO, 1977)) for deaths before 2000, and to ICD-10 (WHO, 1992) for those that occurred from 2000 onward. We also linked annual place of residence (six-character postal code) using Historical Tax Summary Files (Peters et al., 2013). In urban areas, six-character residential postal codes correspond to one side of a city block or to a single apartment building; in rural areas they may represent a larger area. The present study is limited to those respondents who, at enrollment, were living within 5 km of a provincial monitoring site where PM_{2.5} oxidative potential was measured (described below). This criterion was meant to reduce potential exposure measurement error for PM_{2.5} oxidative potential as spatial variations in oxidative potential have been shown to exceed those for mass concentrations (Yang et al., 2015b). Additionally, we excluded recent

Table 1
Descriptive statistics for the study cohort.

Status at baseline		Subjects		Deaths ^a	
		n ^b	%	n ^b	%
All		193,300		40,300	
Sex	Men	93,800	48.5	21,800	54.1
	Women	99,600	51.5	18,500	45.9
Age	25–34	49,700	25.7	800	2.0
	35–64	106,000	54.8	14,200	35.2
	65–74	24,600	12.7	14,000	34.8
	75–89	13,100	6.8	11,300	28.0
Aboriginal status	Yes	2800	1.4	400	0.9
	No	190,600	98.6	39,900	99.1
Visible minority	Yes	17,300	9.0	1800	4.5
	No	176,000	91.0	38,400	95.5
Marital status	Divorced/separated/ widowed	35,400	18.3	12,800	31.9
	Married/common-law	125,400	64.9	23,900	59.3
	Single	32,500	16.8	3600	8.9
Income quintile	Lowest	28,700	14.9	9700	24.2
	Lower middle	33,600	17.4	8800	21.8
	Middle	39,700	20.6	7400	18.4
	Upper Middle	43,900	22.7	7100	17.7
Education	Upper	47,300	24.5	7200	17.9
	< High school graduation	63,000	32.6	21,200	52.7
	High school graduation with/without trade cert.	66,600	34.5	12,200	30.3
	Some postsecondary or college diploma	30,000	15.5	3600	9.1
	University degree	33,700	17.4	3200	7.9
Occupational class	Not applicable	51,000	26.4	26,500	65.8
	Management	16,800	8.7	1600	3.9
	Professional	25,300	13.1	1800	4.5
	Technical	40,700	21.1	4000	9.9
	Semi-skilled	45,100	23.3	4500	11.2
	Unskilled	14,300	7.4	1900	4.7
Labor force status	Employed	125,600	65.0	10,900	27.2
	Unemployed	9100	4.7	1000	2.4
	Not in labor force	58,600	30.3	28,400	70.5
Immigrant status	Canadian born	131,100	67.8	26,500	65.8
	Immigrant	62,200	32.2	13,800	34.2

^a Non-accidental deaths.

^b Subject and deaths counts rounded to nearest 100 for confidentiality; percentages based on original values. SD, standard deviation.

immigrants (those immigrating in the eight years prior to baseline) given that they tend to have notably different health behaviors and outcomes (Ng, 2011) and may have different past exposures than the native-born Canadian population. Use of CanCHEC for population-based research was approved by the Statistics Canada Policy Committee after consultation with Statistics Canada Confidentiality and Legislation Committee, Data Access and Control Services Division, and the Federal Privacy Commissioner. This approval is equivalent to that of standard research ethics boards.

2.2. PM_{2.5} mass concentrations

Ground monitoring data were compiled for annual average PM_{2.5} mass concentrations for the 30 sites used to assess regional PM_{2.5} oxidative potential (described below). Historical data were available beginning in 1998 but the number of years available differed between sites (mean=7.7 years; range: 3–12 years). All PM_{2.5} data were collected using Thermo TEOM 1400AB monitors at provincial monitoring sites and were adjusted for potential bias during winter months (owing to the loss of volatile components) using data provided by Environment Canada. All participants residing within 5 km of a given monitoring site at enrollment were assigned the same mean PM_{2.5} concentration from ground monitors (for the years 1998–2009).

2.3. $PM_{2.5}$ oxidative potential

Regional $PM_{2.5}$ samples were collected between 2012 and 2013 from 30 provincial monitoring sites located across Ontario. Sites were located mainly in urban areas, although several rural sites were also included (populations less than 20,000); the province of Ontario uses Thermo TEOM 1400AB monitors with Sample Equilibration Systems across its network at an operating temperature of 30 °C. Typically, $PM_{2.5}$ filters from these monitors are changed and discarded every 4–6 weeks as part of normal operating procedures. For this study, station managers were asked to keep the used filters and to store them in sealed containers (petri dishes inside sealed zip-lock bags); otherwise stations functioned normally and did not alter the timing of filter change out. Station managers also completed log sheets to record the start and end dates/times of filter collection, the mean flow rate (typically 3 l/min), and the mean $PM_{2.5}$ concentration ($\mu\text{g}/\text{m}^3$) over the collection period. These filters were subsequently sent for oxidative potential analysis (described below). Multiple $PM_{2.5}$ filters were collected from most sites (range: 1–7); as a result oxidative potential values were estimated using a time-weighted average over the entire monitoring period in each region.

2.4. $PM_{2.5}$ oxidative potential characterization

Regional $PM_{2.5}$ filters were equilibrated for 24 h in a weighing room with temperature (18–22 °C) and humidity (45–50%) controls. Filters were weighed pre- and post- extraction using a Sartorius model MC5 microbalance. PM was extracted from the filter samples into HPLC grade methanol through sonication with a titanium probe sonicator operated at an amplitude of 5 μm for 30 s. Filters were then rinsed with additional methanol and sample extracts were dried with a gentle stream of nitrogen gas while in a water bath heated to 37 °C. Samples were re-suspended in a 5% methanol/ultrapure water solution to 75 $\mu\text{g}/\text{mL}$. The extraction efficiency for each filter sample was calculated by comparing station-reported filter masses with the net loss of filter mass from the filters, pre- to post-extraction. The median extraction efficiency was 97%.

Re-suspended $PM_{2.5}$ samples were incubated with a synthetic human respiratory tract lining fluid for 4 h at 37 °C (Godri et al., 2011; Mudway et al., 2004). This fluid was a 200 μM composite solution of physiologically-relevant antioxidants including ascorbate (AA), urate, and glutathione (GSH). In this study, the term *oxidative potential* (i.e. OP^{GSH} and OP^{AA}) is used to describe the ability of regional $PM_{2.5}$ filter extracts to deplete antioxidants in the simulated respiratory tract lining fluid (% depletion/ μg) whereas *oxidative burden* is used to describe the product of $PM_{2.5}$ mass concentrations and regional oxidative potential measurements (i.e. $PM_{2.5} * OP^{\text{GSH}}$ or $PM_{2.5} * OP^{\text{AA}}$).

Glutathione and ascorbate depletion were used to measure regional $PM_{2.5}$ oxidative potential expressed on a per mass basis (% depletion/ μg); urate was not included as previous evidence suggests that PM does not have an important impact on urate depletion (Künzli et al., 2006; Mudway et al., 2004; Zielinski et al., 1999). The plate was incubated at 37 °C in a UV-vis plate reader (Molecular Devices, SpectraMax 190). Ascorbate-related oxidative potential (OP^{AA}) was measured using absorbance spectra and percent ascorbate depletion was measured as the percent change in absorbance at a wavelength of 260 nm over the 4-h incubation period.

Following the 4-h incubation period, the remaining reduced glutathione concentration was determined using the oxidized glutathione -reductase-5,5'-dithio-bis(2-nitrobenzoic acid) recycling assay (Baker et al., 1990). Glutathione-related oxidative potential (OP^{GSH}) was expressed as the percent change between

each sample and a 4-h incubated water blank. $PM_{2.5}$ samples and controls were analyzed in triplicate on a 96-well plate at a final concentration of 75 $\mu\text{g}/\text{mL}$. Known positive (non-ferrous dust, NRC PD-1) and negative (carbon black, Arosperse 15B) controls were run in parallel with $PM_{2.5}$ samples to evaluate inter-experimental standardization (Supplemental material Fig. S1). The stability of the oxidative potential metric is dependent on the PM component(s) mediating antioxidant depletion. Given the analyzed filters were collected from TEOM instruments fitted with heated inlets, only non-volatile chemical species would be retained on the filter. Assessment of non-volatile PM components is expected to yield stable oxidative potential results (Godri et al., 2010).

Separate exposure metrics were generated for glutathione ($PM_{2.5} * OP^{\text{GSH}}$) and ascorbate ($PM_{2.5} * OP^{\text{AA}}$)-related oxidative burden by multiplying $PM_{2.5}$ mass concentrations assigned to cohort members by regional estimates of mean oxidative potential. Associations between $PM_{2.5} * OP^{\text{GSH}}$, $PM_{2.5} * OP^{\text{AA}}$, and cause-specific mortality were evaluated separately in Cox proportional hazard models (described below).

2.5. Statistical analysis

We used Cox proportional hazards models to estimate associations between air pollution and mortality. We estimated hazard ratios (HRs) stratified by sex and by five-year age groups from age 25 to 89; analyses were restricted to subjects less than 90 years of age due to potential inaccuracies in record-linkages among older subjects (e.g., the address reported on the annual income tax filings of older subjects may reflect those of next of kin, or of institutional facilities). Subjects were censored at time of death or if they were lost to follow-up due to end of study period or lack of postal code information.

We adjusted our survival models for aboriginal ancestry, visible minority status, immigrant status, marital status, highest level of education, employment status, occupational classification, and quintiles of household income (see Table 1 for coding). We developed models for five causes of death including: all non-accidental causes (ICD-9: < 800; ICD-10: A-R), lung cancer (ICD-9: 162; ICD-10: C33–C34), cardio-metabolic diseases (i.e., circulatory plus diabetes) (ICD-9: 390–459, 250; ICD-10: I00–I99; E10–E14), ischemic heart disease (ICD-9: 410–414; ICD-10: I20–I25), and diseases of the respiratory system (ICD-9: 460–519; ICD-10: J00–J46). All hazard ratios reflect interquartile range (IQR) changes in exposure. As sensitivity analyses, we evaluated the impact of oxidative burden measures on mortality risks excluding sites below the 10th percentile of total days monitored (< 33 days) for oxidative potential (Supplemental material Fig. S2). In addition, two pollutant models were examined including both $PM_{2.5}$ and OP^{GSH} or OP^{AA} in the model to evaluate the independent effects of particle mass concentrations and oxidative potential on cause-specific mortality.

Information on smoking status and obesity, which are important risk factors for cardiorespiratory mortality, were not collected as part of the 1991 census. As a result, we did not have personal-level information on these variables for cohort members. Therefore, we used a method to mathematically adjust our HRs by examining the relationship between these variables and air pollution in a secondary dataset (i.e., the nationally representative Canadian Community Health survey (CCHS)), which did include information on smoking and obesity (as well as the other variables included in our survival models). Occupational class was not included in the adjusted estimates since this variable is not measured in all cycles of the CCHS. With this method we were able to adjust our observed hazard ratios for risk factors not reported in our dataset (e.g. smoking and obesity) while simultaneously

controlling for the risk factors that were included in our models (e.g., education, income). The details of this method have been reported previously (Shin et al., 2014) and the method has been implemented in several other studies using CanCHEC (Crouse et al., 2015) and other cohorts (Chen et al., 2013; Villeneuve et al., 2013). For the present study the indirect adjustment was based on 68,670 CCHS subjects (aggregated from CCHS survey years 2001–2008), all of whom were less than 90 years of age (same as our main cohort) and lived within 5 km of one of the 30 stations where oxidative potential was measured. All statistical analyses were conducted using SAS version 9.3 and R version 3.1.1.

3. Results

In total, 40,300 non-accidental deaths occurred over the follow-up period including 16,100 deaths from cardiometabolic causes, 3000 for respiratory causes (excluding lung cancer), and 3200 for lung cancer. The characteristics of cohort members in this study were similar to those of the larger national cohort (Crouse et al., 2012). In particular, cohort members were predominantly Canadian-born, employed, married/common law, and were not visible minorities or of Aboriginal ancestry (note: subject counts are rounded to the nearest 100 in order to comply with Statistics Canada confidentiality requirements) (Table 1).

On average, regional $PM_{2.5}$ oxidative potential analyses reflected 132 days per site (median: 120 days; interquartile range: 71–204 days). Glutathione ($r=0.08$) and ascorbate-related oxidative potential ($r=-0.012$) were not correlated with concurrent $PM_{2.5}$ mass concentrations at provincial monitoring sites or the

Table 2

Descriptive statistics for ambient $PM_{2.5}$ and $PM_{2.5}$ -oxidative burden.

Exposure	Mean (SD)	Median (range)	IQR
<i>Regional oxidative potential</i>			
Op ^{GSH} (% Depletion/ μ g)	0.15 (0.07)	0.15 (0.01–0.34)	0.12
Op ^{AA} (% Depletion/ μ g)	0.26 (0.08)	0.24 (0.05–0.39)	0.12
<i>$PM_{2.5}$ and $PM_{2.5}$ oxidative burden</i>			
$PM_{2.5}$ (μ g/ m^3)	9.81 (1.59)	10.09 (4.74–13.62)	2.17
$PM_{2.5}$ *Op ^{GSH}	1.46 (0.72)	1.50 (0.05–2.88)	1.28
$PM_{2.5}$ *Op ^{AA}	2.54 (0.91)	2.42 (0.44–4.15)	1.02

Op^{GSH}, glutathione-related oxidative potential; Op^{AA}, ascorbate-related oxidative potential; $PM_{2.5}$ *Op^{GSH}, glutathione-related oxidative burden; $PM_{2.5}$ *Op^{AA}, ascorbate-related oxidative burden; IQR, interquartile range.

number of days sampled at a given site ($r \leq 0.14$). In addition, these two measures were weakly correlated with each other ($r=0.14$). Mean $PM_{2.5}$ and oxidative potential values are shown for each site in Fig. 1. Mean coefficients of variation (standard deviation/mean) for glutathione and ascorbate-related oxidative potential were 0.65 (IQR: 0.28–0.87) and 0.39 (IQR: 0.19–0.58), respectively. However, the opportunistic nature of filter collection was not ideally suited to characterizing temporal variations in oxidative potential as each sample did not reflect the same number of days.

$PM_{2.5}$ mass concentration exposures assigned to cohort members (Table 2) were moderately correlated ($r < 0.54$) with glutathione-related oxidative burden ($PM_{2.5}$ *Op^{GSH}) and ascorbate-related oxidative burden ($PM_{2.5}$ *Op^{AA}) whereas glutathione and ascorbate-related oxidative burden were weakly correlated with each other ($r=0.27$). Hazard ratios describing the relationship between cause-specific mortality, $PM_{2.5}$, and $PM_{2.5}$ oxidative

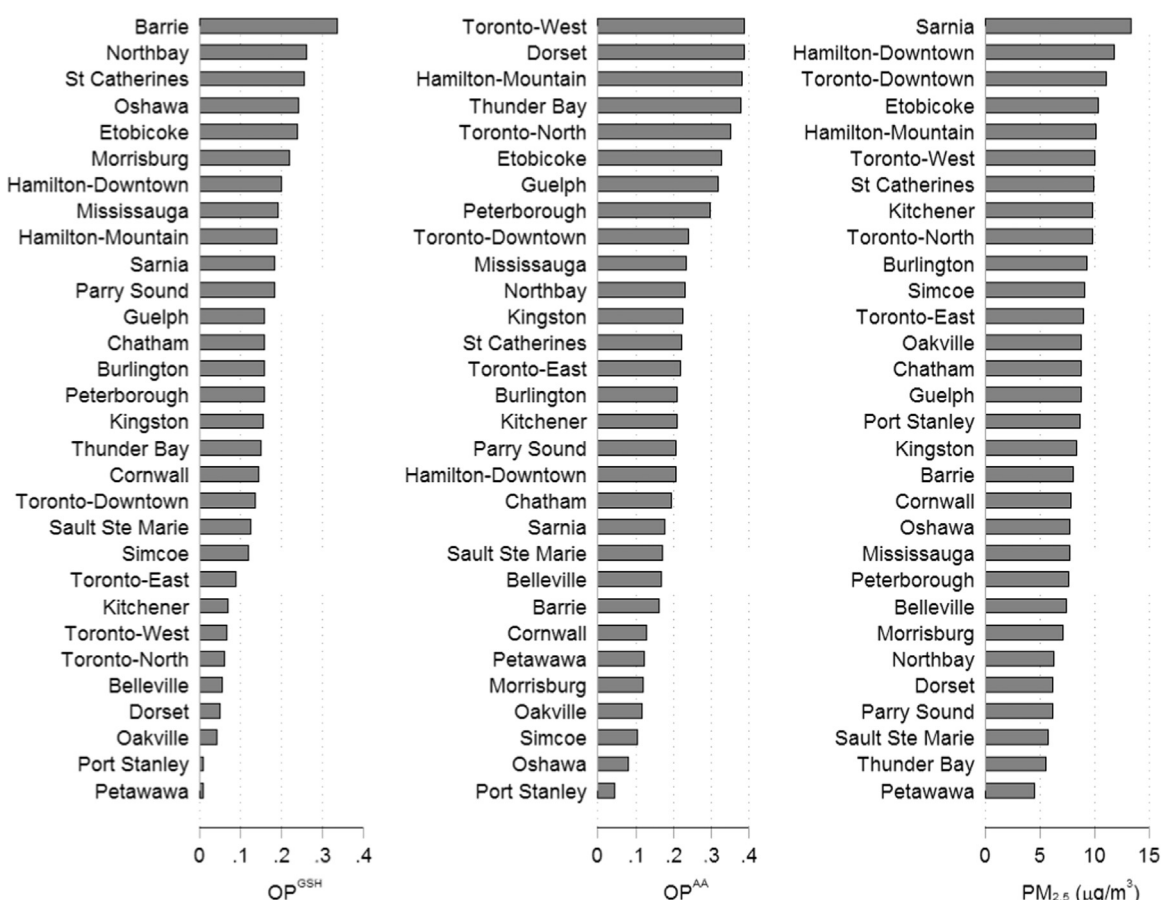


Fig. 1. Mean $PM_{2.5}$ mass concentrations and glutathione (Op^{GSH}) and ascorbate-related (Op^{AA}) oxidative potential (% depletion/ μ g) across sites in Ontario, Canada.

Table 3
Hazard ratios (95% CIs) per interquartile change in PM_{2.5} and PM_{2.5} oxidative burden by cause of death.

Cause of Death	Deaths (n)	PM _{2.5}	Oxidative burden measures	
			PM _{2.5} *OP ^{GSH}	PM _{2.5} *OP ^{AA}
		HR (95% CI)	HR (95% CI)	HR (95% CI)
Non-accidental	40,300	1.026 (1.012, 1.039)	1.029 (1.012, 1.047)	1.000 (0.989, 1.012)
Lung cancer	3200	1.050 (1.001, 1.100)	1.117 (1.052, 1.186)	0.970 (0.933, 1.009)
Respiratory disease	3000	1.033 (0.984, 1.085)	1.045 (0.983, 1.112)	1.000 (0.960, 1.042)
Cardio-metabolic	16,100	1.015 (0.993, 1.036)	1.029 (1.002, 1.057)	0.970 (0.953, 0.988)
Ischemic heart disease	8600	0.989 (0.961, 1.018)	1.007 (0.971, 1.045)	0.946 (0.924, 0.969)

All models are adjusted for age, sex, aboriginal ancestry, visible minority status, immigrant status, marital status, highest level of education, employment status, occupational classification, and household income.

burden are shown in Table 3. Each IQR increase in ambient PM_{2.5} was associated with increased risks of non-accidental (HR=1.026, 95% CI: 1.012, 1.039) and lung cancer mortality (HR=1.050, 95% CI: 1.001, 1.100) but risk estimates for other outcomes were not significantly increased.

Glutathione-related oxidative burden (PM_{2.5}*OP^{GSH}) was associated with non-accidental mortality as well as lung cancer and cardiometabolic mortality (Table 3). The largest change was observed for lung cancer mortality as magnitude of the hazard ratio for glutathione-related oxidative burden was more than double that observed for PM_{2.5}. Moreover, model fit was improved for glutathione-related oxidative potential (AIC=55,598) compared to PM_{2.5} (AIC=55,608) in lung cancer models (AIC values for other outcomes were similar). Excluding sites below the 10th percentile of total monitoring days for oxidative potential did not meaningfully change the results for glutathione-related oxidative burden (PM_{2.5}*OP^{GSH}) Supplemental material Table S3). Ascorbate-related oxidative burden was not associated with increased risks of non-accidental or cause-specific mortality.

The results of two pollutant models including both PM_{2.5} and OP^{GSH} are summarized in Table 4. PM_{2.5} and OP^{GSH} were each independently associated with lung cancer mortality although the magnitude of association was greater for OP^{GSH}. In particular, each inter-quartile range increase in OP^{GSH} was associated with a 9.1% (95% CI: 2.7, 16) increased risk of lung cancer mortality whereas a 4.9% (95% CI: 0.1, 9.9) increase was observed for PM_{2.5}. Glutathione-related oxidative potential was not associated with any other mortality outcomes in two-pollutant models but PM_{2.5} was associated with non-accidental mortality after adjusting for OP^{GSH}.

PM_{2.5}, OP^{GSH}, and glutathione-related oxidative burden were all weakly associated ($r \leq 0.05$) with smoking and obesity in the CCHS data used for the indirect adjustment. Applying the indirect

adjustment for cigarette smoking and obesity generally resulted in small decreases (~1%) in the magnitudes of associations between PM_{2.5}, PM_{2.5} oxidative burden, and cause specific mortality and precision was also decreased (Supplemental material Table S4). In particular, PM_{2.5} was not associated with lung cancer mortality after indirect adjustment for smoking and obesity (HR=1.037, 95% CI: 0.986, 1.09); the hazard ratio for glutathione-related oxidative burden (PM_{2.5}*OP^{GSH}) was also decreased but remained significantly elevated (HR=1.073, 95% CI: 1.005, 1.146). A similar pattern was observed in two-pollutant models for PM_{2.5} and OP^{GSH} as the magnitude and precision of observed associations generally decreased with indirect adjustment for smoking and obesity (Supplemental material Table S5) and OP^{GSH} was no longer associated with lung cancer mortality (HR=1.051, 95% CI: 0.984, 1.122). The relationship between lung cancer mortality and glutathione-related oxidative burden (with indirect adjustment for smoking and obesity) is shown in Fig. 2.

4. Discussion

To our knowledge this is the first study to examine the association between PM_{2.5} oxidative burden and cause specific mortality. In general, our findings suggest that glutathione-related oxidative burden (PM_{2.5}*OP^{GSH}) may be more strongly associated

Table 4
Hazard ratios (95% CIs) for two-pollutant models per interquartile change in exposure by cause of death.

Cause of death	Deaths (n)	Exposure	HR (95% CI)
Non-accidental	40,300	PM _{2.5}	1.025 (1.012, 1.039)
		OP ^{GSH}	1.016 (0.999, 1.034)
Lung cancer	3200	PM _{2.5}	1.049 (1.001, 1.099)
		OP ^{GSH}	1.091 (1.027, 1.158)
Respiratory disease	3000	PM _{2.5}	1.033 (0.984, 1.084)
		OP ^{GSH}	1.034 (0.972, 1.101)
Cardio-metabolic	16,100	PM _{2.5}	1.014 (0.993, 1.036)
		OP ^{GSH}	1.023 (0.995, 1.051)
Ischemic heart disease	8600	PM _{2.5}	0.989 (0.961, 1.018)
		OP ^{GSH}	1.014 (0.977, 1.052)

OP^{GSH}, glutathione-related oxidative potential. All models are stratified by age and sex, and adjusted for Aboriginal status, visible minority status, immigrant status, marital status, highest level of education, employment status, occupational classification, and household income quintile

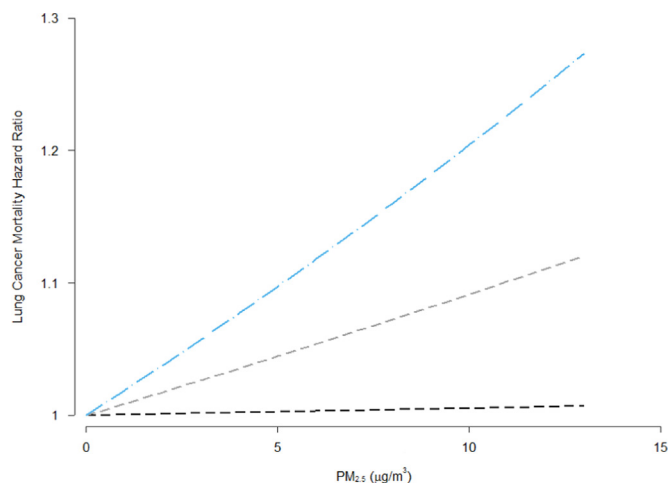


Fig. 2. Relationship between glutathione-related oxidative burden (PM_{2.5}*OP^{GSH}) and lung cancer mortality. The upper and lower lines indicate relationships at the maximum and minimum values for glutathione-related oxidative potential (OP^{GSH}), respectively. The middle line reflects the relationship at the median value of OP^{GSH}. Models are adjusted for age, sex, aboriginal ancestry, visible minority status, immigrant status, marital status, highest level of education, employment status, occupational classification, and household income with indirect adjustment for smoking and obesity.

with lung cancer mortality than $PM_{2.5}$ and that spatial variations in $PM_{2.5}$ oxidative burden exceed spatial gradients in $PM_{2.5}$ mass concentrations. This finding is particularly interesting as it may explain in part why it is possible to detect $PM_{2.5}$ -mortality associations at very low mass concentrations: a change in exposure on the mass concentration scale may translate into much larger changes on the scale of oxidative burden, which ultimately may be more relevant to health. In addition, this variation may also explain heterogeneity between studies, as stronger associations may occur in regions where the spatial pattern of $PM_{2.5}$ more closely reflects spatial patterns in oxidative potential. In general, the fact that glutathione-related oxidative burden was associated with cause-specific mortality is important because this measure may allow regulators to evaluate health risks across regions with similar $PM_{2.5}$ mass concentrations that would otherwise be assumed to be equally hazardous.

Ambient particulate air pollution is classified as a known human carcinogen based largely on epidemiological evidence related to lung cancer (Hamra et al., 2014). In this study, glutathione-related oxidative burden was more strongly associated with lung cancer mortality than $PM_{2.5}$ mass concentrations. The biological plausibility of this finding is supported by the role of glutathione in the body as well as the fact that inhaled particles are thought to increase lung cancer risk through their ability to cause oxidative stress (Knappen et al., 2004). In particular, glutathione is known to play an important role in oxidant defense and counteracts oxidative damage (e.g. DNA damage) and inflammatory responses caused by reactive oxygen species (Rahman and MacNee, 2000, 1999; Pastore et al., 2003; Valko et al., 2007; Ballatori et al., 2009). Moreover, disorders in enzyme functions related to glutathione have been associated with cancer development (Pastore et al., 2003), and for lung cancer specifically, existing evidence suggests that GSTM1 null mutations (which reduce anti-oxidant capacity) may increase lung cancer risk (Carlsten et al., 2008). Other studies of $PM_{2.5}$ oxidative burden and cause-specific mortality were not identified; however, it seems that glutathione-related oxidative burden may be more strongly associated with lung cancer mortality because this metric may be more closely related to the underlying biological mechanisms linking exposure to disease onset.

Exposure to particulate air pollution is known to be associated with cardiorespiratory morbidity; however, isolating the biological mechanisms explaining these relationships is challenging as studies generally rely on mass concentrations for a given particle size fraction despite the fact that much of this mass may be biologically inert (Borm et al., 2007). Indeed, there is a growing consensus that only a small fraction of particulate matter deposited in the lung can cause adverse health effects, most commonly through oxidative stress pathways (Araujo and Nel, 2009; Li et al., 2003; Mudway et al., 2004; Squadrito et al., 2001). In this study, we characterized $PM_{2.5}$ oxidative potential using a metric based on the ability of $PM_{2.5}$ extracts to deplete antioxidants in a synthetic respiratory tract lining fluid. This antioxidant coating is the body's first defense against airborne pollutants and thus this metric provides an indication of the toxicity of PM by measuring its capacity to deplete physiologically relevant antioxidants. While previous studies have also used this assay to evaluate the oxidative potential of PM (Godri et al., 2010; Kelly et al., 2011; Kurmi et al., 2013; Mudway et al., 2004), it is important to note that there is no "gold-standard" measure of $PM_{2.5}$ oxidative potential and other methods are available to evaluate this metric (Ayres et al., 2008). In general, measures collected in this study likely reflect metal-dependent oxidative potential owing to the potential loss of the volatile/semi-volatile fraction of PM with TEOM monitors (Mudway, 2004). In addition, the assay does not capture oxidative potential caused by biotransformation *in vivo*. Therefore, regional oxidative potential measures determined in this study likely

underestimate the total oxidative burden of PM in a given area.

While oxidative stress is thought to play an important role in air pollution impacts on cardiovascular health (Araujo and Nel, 2009; Weichenthal et al., 2013), only small changes in hazard ratios were observed for cardiovascular outcomes in association with glutathione-related oxidative burden. The reason for this is not entirely clear, although one explanation may be that the assay used to measure oxidative potential does not adequately capture the process through which $PM_{2.5}$ contributes to oxidative stress in the vasculature. Indeed, one other study used the same assay to evaluate the impact of PM_{10} oxidative potential on carotid intima-media thickness and did not observe a stronger effect compared with PM_{10} (Tonne et al., 2012). Moreover, ascorbate-related oxidative burden was not associated with any of the outcomes examined and was a worse predictor of cause-specific mortality than $PM_{2.5}$ alone. Therefore, while ascorbate is known to be an important anti-oxidant (Valko et al., 2007), ascorbate depletion may not play a major role in explaining the association between particulate air pollution and cardiorespiratory mortality.

None of the exposure measures were significantly associated with respiratory mortality although hazard ratios for $PM_{2.5}$ and glutathione-related oxidative burden were similar in magnitude to previous studies of long-term exposure to $PM_{2.5}$ and respiratory mortality (Laden et al., 2006; Beelen et al., 2008; Katanoda et al., 2011). Likewise, $PM_{2.5}$ was not associated with ischemic heart disease mortality as previously reported for the CanCHEC cohort as a whole (Crouse et al., 2012). However, the present analyses were limited to a subset of the CanCHEC cohort and our findings require replication on a national scale once oxidative potential data are available across Canada.

While this study had a number of important advantages including a broad spatial network of oxidative potential measurements in urban and rural areas, a follow-up period of more than 15 years, and adjustment for many socio-demographic risk factors for mortality, it is important to note several limitations. First, our assessment of regional $PM_{2.5}$ oxidative potential was based on samples collected between 2012 and 2013 and thus these estimates were likely imprecise measures of true long-term values that may be most relevant to health. In particular, spatial variations within cities (Janssen et al., 2014; Yanosky et al., 2012; Yang et al., 2015a) as well as changes in source profiles over time likely contributed to exposure measurement error for oxidative potential. Moreover, measurement error for oxidative potential was likely greater than for $PM_{2.5}$ (owing to less spatial variability for $PM_{2.5}$) thus hindering a direct comparison of these two measures. Nevertheless, our estimates of $PM_{2.5}$ oxidative potential did appear to capture at least some useful information as weighting $PM_{2.5}$ estimates increased hazard ratios and improved model fit for some outcomes and this would not be expected if we were simply weighted estimates by random noise. A further limitation was the fact that we did not know which specific components/sources of $PM_{2.5}$ were the most important predictors of oxidative potential. However, previous studies suggest that traffic sources (Janssen et al., 2014; Yanosky et al., 2012; Yang et al., 2015a), biomass burning (Bates et al., 2015; Kurmi et al., 2013) and metal components of $PM_{2.5}$ (Supplemental material Table S6) may be important predictors of oxidative potential and thus these factors may have contributed to spatial differences $PM_{2.5}$ oxidative burden across Ontario. Similarly, a number of other methods are available to characterize the oxidative potential of $PM_{2.5}$ (Ayres et al., 2008) and future studies should explore multiple assays to determine which measures are most strongly associated with adverse health outcomes.

An additional limitation of our study is that our cohort lacked individual-level data on obesity and smoking behavior. To address this limitation, we indirectly adjusted hazard ratios using an established method and this additional adjustment for smoking and obesity did not materially change the message of our findings for

glutathione related oxidative burden with respect to lung cancer mortality; however, associations for non-accidental mortality and cardio-metabolic mortality were no longer statistically significant. In general, we cannot rule out residual confounding by smoking or obesity and further evaluation of the relationship between PM_{2.5} oxidative burden and cause-specific mortality is needed in cohorts with individual-level information on other potentially important confounding factors.

Finally, people do not spend all of their time at their residence, and many subjects likely moved during follow-up. As such, exposures assigned at baseline are almost certainly imperfect measures of long-term exposure and exposure measurement error likely impacted our results. Nevertheless, many previous longitudinal cohorts (based around the world) have made similar assumptions related to residential exposures at baseline and have reported associations between long-term exposures to air pollution and cause-specific mortality (Beelen et al., 2008; Carey et al., 2013; Cesaroni et al., 2013; Crouse et al., 2012; Jerrett et al., 2009, 2013; Krewski et al., 2009; Laden et al., 2006; Raaschou-Nielsen et al., 2012). However, our findings suggest that exposure measurement error from residential mobility may be more important for PM_{2.5} oxidative burden than for PM_{2.5} given that spatial variations in PM_{2.5} oxidative burden greatly exceed those for PM_{2.5}.

5. Conclusion

Glutathione-related oxidative burden may be more strongly associated with lung cancer mortality than PM_{2.5} mass concentrations. Further evaluation of this exposure measure is encouraged in large population-based studies as spatial variations in oxidative potential may exceed spatial differences in ambient PM_{2.5}. In particular, application of this metric may facilitate the evaluation of health risks across regions with similar mass concentrations previously assumed to be equally hazardous.

Competing interests

None declared.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.envres.2015.12.013>.

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